

Revised Problem Set: Density of States and Electron Concentration

Problem 1: Effective Density of States and Electron Concentration (300 K)

Context: To find electron concentration in a semiconductor, we must account for how electronic states are distributed in energy (the density of states) and how likely those states are to be occupied (the Fermi-Dirac distribution).

Problem Statement

A semiconductor has electron effective mass $m_n^* = 0.25 m_0$. At $T = 300$ K, the conduction-band edge is at $E_c = 0$, and the Fermi level is $E_F = -0.25$ eV.

- (a) Compute the effective density of states N_c .
- (b) Compute the electron concentration n using the non-degenerate approximation.

Detailed Solution

Part (a): Effective Density of States

Why we need this:

In the conduction band, the density of electronic states varies with energy near the band edge as $g(E) \propto \sqrt{E - E_c}$. Rather than integrate over all energies, we can define an effective DOS N_c —an effective number of states that would give the same electron count as if all carriers were at the band edge.

The formula:

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2} \quad (1)$$

Physical meaning of components:

- The factor of **2** accounts for spin degeneracy (spin-up and spin-down electrons).
- The term $\frac{2\pi m_n^* k_B T}{h^2}$ is the thermal de Broglie wavelength cubed—it represents the quantum volume occupied by one electron at thermal energy $k_B T$.

- The $3/2$ power comes from integrating over the three-dimensional density of states.
- Higher temperature and larger effective mass increase N_c (more accessible states).

Step-by-step calculation:

Given: $m_n^* = 0.25 m_0$, $T = 300$ K

Physical constants:

$$m_0 = 9.109 \times 10^{-31} \text{ kg}$$

$$k_B = 1.381 \times 10^{-23} \text{ J/K}$$

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

$$k_B T = 1.381 \times 10^{-23} \times 300 = 4.143 \times 10^{-21} \text{ J}$$

Step 1: Calculate m_n^*

$$m_n^* = 0.25 \times 9.109 \times 10^{-31} = 2.277 \times 10^{-31} \text{ kg}$$

Step 2: Calculate $2\pi m_n^* k_B T$

$$\begin{aligned} 2\pi m_n^* k_B T &= 2\pi \times 2.277 \times 10^{-31} \times 4.143 \times 10^{-21} \\ &= 2\pi \times 9.436 \times 10^{-52} \\ &= 5.930 \times 10^{-51} \text{ kg} \cdot \text{J} \end{aligned}$$

Step 3: Calculate h^2

$$h^2 = (6.626 \times 10^{-34})^2 = 4.390 \times 10^{-67} \text{ J}^2 \cdot \text{s}^2$$

Step 4: Calculate the argument of the power

$$\frac{2\pi m_n^* k_B T}{h^2} = \frac{5.930 \times 10^{-51}}{4.390 \times 10^{-67}} = 1.351 \times 10^{16} \text{ m}^{-2}$$

Step 5: Take the $3/2$ power

$$\begin{aligned} (1.351 \times 10^{16})^{3/2} &= (1.351)^{3/2} \times (10^{16})^{3/2} \\ &= 1.568 \times 10^{24} \text{ m}^{-3} \end{aligned}$$

Step 6: Multiply by factor of 2 (spin degeneracy)

$$N_c = 2 \times 1.568 \times 10^{24} = 3.136 \times 10^{24} \text{ m}^{-3}$$

Convert to cm^{-3} (divide by 10^6):

$$\boxed{N_c = 3.14 \times 10^{18} \text{ cm}^{-3}}$$

Alternative expression using reference form:

Often it is easier to use the reference form:

$$N_c(T) = N_{c,0} \left(\frac{T}{300} \right)^{3/2} \quad (2)$$

where $N_{c,0} \approx 2.8 \times 10^{19} \text{ cm}^{-3}$ for $m_n^*/m_0 = 1$ at 300 K.
For $m_n^* = 0.25 m_0$:

$$\begin{aligned} N_c &= 2.8 \times 10^{19} \times (0.25)^{3/2} \\ &= 2.8 \times 10^{19} \times 0.125 \\ &= \boxed{3.5 \times 10^{18} \text{ cm}^{-3}} \end{aligned}$$

The slight difference from direct calculation arises from rounding in reference values.

Part (b): Electron Concentration (Non-degenerate approximation)

Starting point:

The total electron concentration is

$$n = \int_{E_c}^{\infty} g(E) f(E) dE \quad (3)$$

where $g(E)$ is the density of states and $f(E)$ is the Fermi-Dirac distribution.

Non-degenerate approximation:

This applies when $E_c - E_F \gg k_B T$ (Fermi level is well below the band edge). In this case:

- The Fermi-Dirac distribution simplifies: $f(E) \approx e^{-(E-E_F)/k_B T}$ (Boltzmann tail).
- The integral simplifies to:

$$n \approx N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \quad (4)$$

Physical meaning:

This says electron concentration is proportional to the effective DOS times the thermal probability of occupying states at the band edge.

Check condition:

$$E_c - E_F = 0 - (-0.25) = 0.25 \text{ eV vs. } k_B T = 0.0259 \text{ eV}$$

$$\frac{E_c - E_F}{k_B T} = \frac{0.25}{0.0259} = 9.65 \gg 1 \quad \checkmark$$

Non-degenerate approximation is valid.

Calculation:

Using the corrected N_c from Part (a):

$$\begin{aligned} n &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \\ n &= 3.5 \times 10^{18} \times \exp\left(-\frac{0.25}{0.0259}\right) \\ n &= 3.5 \times 10^{18} \times \exp(-9.65) \\ n &= 3.5 \times 10^{18} \times 6.2 \times 10^{-5} \\ n &= \boxed{2.2 \times 10^{14} \text{ cm}^{-3}} \end{aligned}$$

Physical interpretation:

The electron concentration is quite small ($\sim 10^{14} \text{ cm}^{-3}$) because the Fermi level is far below the conduction band edge (by $0.25 \text{ eV} \gg k_B T$). This is typical of a lightly-doped semiconductor or an undoped sample at room temperature. The exponential dependence on $(E_c - E_F)$ shows that the electron concentration is extremely sensitive to Fermi level position—small changes in doping or temperature lead to exponential changes in carrier concentration.

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- (b) Compute the electron concentration n using the non-degenerate approximation.

Detailed Solution

Part (a): Effective Density of States

Why we need this: In the conduction band, the density of electronic states varies with energy near the band edge as $g(E) \propto \sqrt{E - E_c}$. Rather than integrate over all energies, we can define an effective DOS N_c an *effective number of states* that would give the same electron count as if all carriers were at the band edge.

The formula:

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2}$$

Physical meaning of components:

- The factor of **2** accounts for spin degeneracy (spin-up and spin-down electrons).
- The term $\frac{2\pi m_n^* k_B T}{h^2}$ is the **thermal de Broglie wavelength cubed** it represents the quantum volume occupied by one electron at thermal energy $k_B T$.
- The $3/2$ power comes from integrating over the three-dimensional density of states.
- Higher temperature and larger effective mass increase N_c (more accessible states).

Calculation: Given: $m_n^* = 0.25 m_0$, $T = 300$ K, $k_B T = 0.0259$ eV = 4.14×10^{-21} J

$$N_c = 2 \left(\frac{2\pi \times 0.25 \times 9.109 \times 10^{-31} \text{ kg} \times 1.381 \times 10^{-23} \text{ J/K} \times 300 \text{ K}}{(1.055 \times 10^{-34} \text{ J} \cdot \text{s})^2} \right)^{3/2}$$

$N_c \approx 4.0 \times 10^{18} \text{ cm}^{-3}$

Part (b): Electron Concentration (Non-degenerate approximation)

Starting point: The total electron concentration is

$$n = \int_{E_c}^{\infty} g(E) f(E) dE$$

where $g(E)$ is the density of states and $f(E)$ is the Fermi-Dirac distribution.

Non-degenerate approximation: This applies when $E_c - E_F \gg k_B T$ (Fermi level is well below the band edge). In this case:

- The Fermi-Dirac distribution simplifies: $f(E) \approx e^{-(E-E_F)/k_B T}$ (Boltzmann tail).
- The integral simplifies to:

$$n \approx N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

Physical meaning: This says electron concentration is proportional to the effective DOS times the thermal probability of occupying states at the band edge.

Check condition: $E_c - E_F = 0.25 \text{ eV}$ vs. $k_B T = 0.0259 \text{ eV}$, so $\frac{E_c - E_F}{k_B T} \approx 9.6 \gg 1$ ✓ (non-degenerate is valid)

Calculation:

$$\begin{aligned} n &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \\ &= 4.0 \times 10^{18} \times \exp\left(-\frac{0.25}{0.0259}\right) \\ &= 4.0 \times 10^{18} \times e^{-9.65} \\ &= 4.0 \times 10^{18} \times 6.2 \times 10^{-5} \\ \boxed{n \approx 2.5 \times 10^{14} \text{ cm}^{-3}} \end{aligned}$$

Problem 2: Electrons in a Finite Energy Window

Context: Real devices often have interest in specific energy windows (e.g., states accessible to tunneling, or participating in impact ionization). We must integrate the product of DOS and occupancy over that window.

Problem Statement

Using the same material and Fermi level as Problem 1, find the number of electrons per cm^3 occupying states between E_c and $E_c + 0.05 \text{ eV}$ at 300 K.

Detailed Solution

The integral to compute:

$$n_{\Delta E} = \int_{E_c}^{E_c+0.05} g(E) f(E) dE$$

Density of states in conduction band:

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m_n^*}{\hbar^2} \right)^{3/2} \sqrt{E - E_c}$$

Physical meaning: The DOS increases as the square root of energy above the band edge; there are more states at higher energy because the volume of $\mathbf{k}$ -space increases.

Applying the non-degenerate approximation: Since E_F is far below E_c , we have $f(E) \approx e^{-(E-E_F)/k_B T}$ for all energies of interest. Define $\xi = E - E_c$ (energy above band edge). Then:

$$\begin{aligned} n_{\Delta E} &= \int_0^{0.05} \frac{1}{2\pi^2} \left(\frac{2m_n^*}{\hbar^2} \right)^{3/2} \sqrt{\xi} \cdot e^{-(E_c+\xi-E_F)/k_B T} d\xi \\ &= e^{-(E_c-E_F)/k_B T} \int_0^{0.05} \frac{1}{2\pi^2} \left(\frac{2m_n^*}{\hbar^2} \right)^{3/2} \sqrt{\xi} \cdot e^{-\xi/k_B T} d\xi \\ &= N_c e^{-(E_c-E_F)/k_B T} \frac{\int_0^{0.05} e^{-\xi/k_B T} \sqrt{\xi} d\xi}{\int_0^{\infty} e^{-\xi/k_B T} \sqrt{\xi} d\xi} \end{aligned}$$

Interpretation: The ratio of integrals is the *fraction* of all occupied conduction-band electrons lying within the window $[E_c, E_c + 0.05 \text{ eV}]$. It is evaluated numerically using incomplete gamma functions.

Numerical evaluation at 300 K:

$$\frac{\int_0^{0.05} e^{-\xi/k_B T} \sqrt{\xi} d\xi}{\int_0^{\infty} e^{-\xi/k_B T} \sqrt{\xi} d\xi} \approx 0.18$$

Physical meaning: Only about 18% of all conduction-band electrons occupy the lowest 0.05 eV of the bandmost are dispersed to higher energies due to thermal excitation.

Final calculation:

$$n_{\Delta E} \approx 0.18 \times n$$

$$= 0.18 \times 2.5 \times 10^{14}$$

$$\boxed{n_{\Delta E} \approx 4.5 \times 10^{13} \text{ cm}^{-3}}$$

Problem 3: Fermi-Dirac vs. Boltzmann Approximation (300 K)

Context: The Boltzmann approximation is simpler but valid only far from the Fermi level. Understanding when it breaks down is crucial for accurate calculations.

Problem Statement

At $T = 300$ K and with $E_c - E_F = 0.20$ eV, compute $f(E)$ at $E - E_F = 0.05, 0.10, 0.20$ eV using both the exact Fermi-Dirac distribution and the Boltzmann approximation. Comment on accuracy.

Detailed Solution

Fermi-Dirac distribution (exact):

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}$$

Boltzmann approximation (valid when $E \gg E_F$):

$$f_B(E) \approx e^{-(E-E_F)/k_B T}$$

When is Boltzmann valid? When $(E - E_F)/k_B T \gtrsim 3$, the +1 in the denominator of Fermi-Dirac becomes negligible.

Parameter: $k_B T = 0.0259$ eV at 300 K

$E - E_F$ (eV)	$(E - E_F)/k_B T$	$f(E)$ [F-D]	$f_B(E)$ [Boltzmann]	Error (%)
0.05	1.93	0.135	0.147	+9
0.10	3.86	0.0207	0.0214	+3.4
0.20	7.72	4.7×10^{-4}	4.7×10^{-4}	< 0.5

Interpretation:

- At $E - E_F = 0.05$ eV: Boltzmann overestimates by $\sim 9\%$ because it neglects the Pauli exclusion effect at the Fermi level.
- At $E - E_F = 0.10$ eV: Error drops to $\sim 3\%$ (borderline for engineering applications).
- At $E - E_F = 0.20$ eV: Boltzmann is excellent ($< 0.5\%$ error).

Conclusion: The rule of thumb is that Boltzmann is reliable when $(E - E_F) \gtrsim 3k_B T$.

Problem 4: Temperature Dependence of Boltzmann Accuracy (600 K)

Context: As temperature increases, thermal energy becomes comparable to energy differences, and the quantum Fermi-Dirac statistics matter more the Boltzmann approximation becomes less accurate.

Problem Statement

Repeat Problem 3 but at $T = 600$ K with a single energy difference $E - E_F = 0.10$ eV.

Detailed Solution

New thermal energy: $k_B T = 0.0518$ eV (doubled from 300 K)

Ratio: $(E - E_F)/k_B T = 0.10/0.0518 = 1.93$

Fermi-Dirac calculation:

$$\begin{aligned} f(E) &= \frac{1}{1 + e^{1.93}} \\ &= \frac{1}{1 + 6.88} = \frac{1}{7.88} \\ &\approx 0.127 \end{aligned}$$

Boltzmann calculation:

$$f_B(E) \approx e^{-1.93} \approx 0.145$$

Comparison: At 600 K, Boltzmann overestimates by $(0.145 - 0.127)/0.127 \approx 14\%$

Why higher T worsens the approximation:

- At 300 K, $E - E_F = 0.10$ eV was 3.86 times $k_B T$ (Boltzmann valid).
- At 600 K, the same energy difference is only 1.93 times $k_B T$ (comparable to $k_B T$, so quantum effects matter).
- The Pauli exclusion principle (encoded in the +1 term) becomes increasingly important at moderate degeneracy.

Key insight: Higher temperature requires either a larger energy window above E_F or acceptance of larger errors when using Boltzmann.

Problem 5: Fermi Level in n-type Silicon (300 K)

Context: The Fermi level position determines device properties. In doped semiconductors, we use charge neutrality and the relationship between E_F and intrinsic Fermi level E_i to find it.

Problem Statement

Silicon with donor concentration $N_D = 10^{16} \text{ cm}^{-3}$ (complete ionization). Given $N_c = 2.8 \times 10^{19} \text{ cm}^{-3}$, $N_v = 1.04 \times 10^{19} \text{ cm}^{-3}$, $E_g = 1.12 \text{ eV}$. Find $E_F - E_i$.

Detailed Solution

Step 1: Define the intrinsic Fermi level E_i The intrinsic Fermi level is the position in the middle of the gap where electron and hole concentrations are equal. It is defined as:

$$E_i = \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c}$$

For $N_c \approx N_v$, we have $E_i \approx \frac{E_c + E_v}{2}$ (middle of gap).

Step 2: Compute intrinsic carrier concentration n_i By definition:

$$n_i p_i = n_i^2 = N_c N_v \exp\left(-\frac{E_g}{k_B T}\right)$$

$$n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$$

Calculation:

$$\begin{aligned} n_i &= \sqrt{2.8 \times 10^{19} \times 1.04 \times 10^{19}} \times \exp\left(-\frac{1.12}{2 \times 0.0259}\right) \\ &= 1.70 \times 10^{19} \times e^{-21.6} \\ &= 1.70 \times 10^{19} \times 2.8 \times 10^{-10} \\ &\approx 4.8 \times 10^9 \text{ cm}^{-3} \end{aligned}$$

(Note: The problem statement suggests $n_i \approx 1.5 \times 10^{10}$; slight numerical variations arise from rounding.)

Step 3: Use charge neutrality in n-type With complete ionization: $n + N_A^- = p + N_D^+$
Since $N_A = 0$ and all N_D are ionized:

$$n = N_D = 10^{16} \text{ cm}^{-3}$$

Step 4: Relate E_F to E_i using mass action law Starting from:

$$n = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right), \quad p = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

Taking the ratio and using $np = n_i^2$:

$$\frac{n}{p} = \frac{N_c}{N_v} \exp\left(-\frac{E_c - E_v + 2E_F}{k_B T}\right) = \frac{N_c}{N_v} \exp\left(-\frac{E_g + 2E_F}{k_B T}\right)$$

After manipulation, this yields:

$$E_F = E_i + \frac{k_B T}{2} \ln \frac{n}{n_i}$$

Physical meaning: The Fermi level shifts above E_i by an amount proportional to $\ln(n/n_i)$ the *doping ratio*.

Calculation:

$$\begin{aligned} E_F - E_i &= \frac{k_B T}{2} \ln \frac{N_D}{n_i} \\ &= \frac{0.0259}{2} \ln \frac{10^{16}}{4.8 \times 10^9} \\ &= 0.01295 \times \ln(2.08 \times 10^6) \\ &= 0.01295 \times 14.55 \end{aligned}$$

$$\boxed{E_F - E_i \approx 0.188 \text{ eV}}$$

Problem 6: Fermi Level in p-type Silicon (300 K)

Context: In p-type material, holes dominate; the Fermi level shifts downward (toward E_v) relative to E_i , with a shift proportional to $\ln(p/n_i)$.

Problem Statement

For $N_A = 5 \times 10^{17} \text{ cm}^{-3}$ (complete ionization), find $E_i - E_F$ at 300 K using the same N_c, N_v, E_g as Problem 5.

Detailed Solution

Step 1: Charge neutrality in p-type With complete ionization and no donors:

$$p + N_D^+ = n + N_A^-$$

Since $N_D = 0$ and all N_A are ionized:

$$p = N_A = 5 \times 10^{17} \text{ cm}^{-3}$$

Step 2: Fermi level relation for holes By analogous reasoning to Problem 5:

$$E_i - E_F = \frac{k_B T}{2} \ln \frac{p}{n_i}$$

Physical meaning: The Fermi level shifts *below* E_i (hence $E_i - E_F > 0$) when the material is p-doped. The shift magnitude is again proportional to $\ln(\text{doping}/n_i)$.

Calculation:

$$\begin{aligned} E_i - E_F &= \frac{0.0259}{2} \ln \frac{5 \times 10^{17}}{1.5 \times 10^{10}} \\ &= 0.01295 \times \ln(3.33 \times 10^7) \\ &= 0.01295 \times 17.32 \end{aligned}$$

$$\boxed{E_i - E_F \approx 0.224 \text{ eV}}$$

Comparison with Problem 5:

- n-type shift: $E_F - E_i \approx 0.19 \text{ eV}$ (upward)
- p-type shift: $E_i - E_F \approx 0.22 \text{ eV}$ (downward)

The p-type shift is slightly larger because $N_A > N_D$ (stronger doping), pushing the Fermi level further from the intrinsic point.

Summary of Key Physical Insights

1. **Density of States:** N_c quantifies available conduction-band states; it scales as $(m_n^*T)^{3/2}$.
2. **Non-degenerate approximation:** Valid when $|E_F - E_c| \gg k_B T$; enables simple exponential formulas for carrier concentration.
3. **Boltzmann vs. Fermi-Dirac:** Boltzmann is excellent for $E - E_F \gtrsim 3k_B T$ but fails near the Fermi level; accuracy degrades at higher temperature.
4. **Intrinsic Fermi level E_i :** The reference energy where undoped material has equal n and p ; depends slightly on N_c and N_v ratio.
5. **Doped semiconductors:** The Fermi level position is uniquely determined by charge neutrality and the mass-action law; it shifts logarithmically with doping concentration and inversely with temperature.

Problem 7: Donor Ionization at 200 K (Freeze-Out Onset)

Context: As temperature decreases, fewer carriers have enough thermal energy to ionize shallow donors. The ionized fraction transitions from nearly 100% (extrinsic regime) toward freeze-out, where most donors remain neutral. This problem demonstrates the onset of freeze-out and requires solving charge neutrality *simultaneously* with donor statistics.

Problem Statement

An n-type sample has $N_D = 10^{17} \text{ cm}^{-3}$ with shallow donors at $E_c - E_D = 0.025 \text{ eV}$. At $T = 200 \text{ K}$, estimate the ionized fraction N_D^+/N_D . Use $N_c(T) = 2.8 \times 10^{19} (T/300)^{3/2} \text{ cm}^{-3}$ and degeneracy factor $g_D = 1/2$.

Detailed Solution

Physical background: Donors contribute electrons to the conduction band via ionization: $D^0 \rightarrow D^+ + e^-$. The occupancy of donor states is given by the Fermi-Dirac distribution at the donor level. The fraction of ionized donors is:

$$\frac{N_D^+}{N_D} = \frac{1}{1 + g_D e^{(E_F - E_D)/k_B T}}$$

where g_D is the *degeneracy factor* (typically $1/2$ for donors, accounting for spin and orbital degeneracy of the donor ground state).

Step 1: Compute Thermal Energy and Effective DOS

Thermal energy:

$$k_B T = 0.0259 \text{ eV} \times \frac{200}{300} = 0.0173 \text{ eV}$$

Effective DOS at 200 K:

$$N_c(200\text{ K}) = 2.8 \times 10^{19} \left(\frac{200}{300} \right)^{3/2} = 2.8 \times 10^{19} \times 0.544 \approx 1.52 \times 10^{19} \text{ cm}^{-3}$$

Physical meaning: Lower temperature reduces N_c because the thermal de Broglie wavelength increases (electrons occupy more space), effectively reducing the density of *accessible* states.

Step 2: Charge Neutrality Condition

Statement: Assuming no acceptors and complete ionization at 200 K is not yet frozen out:

$$n = N_D^+$$

Physical meaning: Every ionized donor contributes one free electron. Charge neutrality requires the electron concentration equal the ionized donor concentration (electrons from donors are the only significant source).

Step 3: Electron Concentration from Band Statistics

Formula: Using the non-degenerate approximation (valid if $E_c - E_F \gg k_B T$):

$$n = N_c \exp \left(-\frac{E_c - E_F}{k_B T} \right)$$

Define $\Delta = E_c - E_F$ (Fermi level depth). Then:

$$n = N_c e^{-\Delta/k_B T}$$

Step 4: Donor Ionization Statistics

Formula: The fraction of ionized donors is:

$$N_D^+ = \frac{N_D}{1 + g_D e^{(E_F - E_D)/k_B T}}$$

Rewrite in terms of Δ : Note that $E_F - E_D = E_F - E_c + E_c - E_D = -\Delta + (E_c - E_D)$. Since $E_c - E_D = 0.025 \text{ eV}$:

$$N_D^+ = \frac{N_D}{1 + g_D \exp \left[\frac{(E_c - E_D) - \Delta}{k_B T} \right]} = \frac{N_D}{1 + \frac{1}{2} e^{(0.025 - \Delta)/k_B T}}$$

Physical meaning: As Δ increases (Fermi level gets further from band edge), the exponent becomes more negative, and N_D^+ decreases more donors freeze out.

Step 5: Solve the Coupled Equations

System to solve: From charge neutrality and band statistics:

$$N_c e^{-\Delta/k_B T} = \frac{N_D}{1 + \frac{1}{2} e^{(0.025 - \Delta)/k_B T}}$$

Substitute values:

$$1.52 \times 10^{19} \times e^{-\Delta/0.0173} = \frac{10^{17}}{1 + 0.5 \exp \left[\frac{0.025 - \Delta}{0.0173} \right]}$$

Solution method: This transcendental equation is solved by *fixed-point iteration*:

1. Guess $\Delta_0 = 0.05 \text{ eV}$
2. Compute right-hand side: $N_D^+ = N_D / \{1 + 0.5 \exp[(0.025 - \Delta)/0.0173]\}$
3. Compute left-hand side: $n_{\text{calc}} = N_c e^{-\Delta/0.0173}$
4. Adjust Δ until convergence ($|n_{\text{calc}} - N_D^+| < \epsilon$)

Numerical result:

$$\Delta \approx 0.087 \text{ eV}$$

Step 6: Compute Ionized Fraction

$$\begin{aligned} N_D^+ &= \frac{10^{17}}{1 + 0.5 \exp \left[\frac{0.025 - 0.087}{0.0173} \right]} = \frac{10^{17}}{1 + 0.5 \times e^{-3.59}} \\ &= \frac{10^{17}}{1 + 0.5 \times 0.0273} = \frac{10^{17}}{1.0137} \approx 9.9 \times 10^{16} \text{ cm}^{-3} \end{aligned}$$

$$\boxed{\frac{N_D^+}{N_D} \approx 0.99}$$

Interpretation: Even at 200 K, ionization remains $\sim 99\%$ for these very shallow donors ($E_c - E_D = 0.025 \text{ eV}$). The freeze-out regime (where $N_D^+/N_D \ll 1$) requires even lower temperatures or deeper donors. This demonstrates that *extrinsic ionization is robust* at modest cooling, but becomes critical in the freeze-out region where device performance degrades.

Problem 8: Acceptor Ionization at 150 K (Freeze-Out Regime)

Context: Acceptors ionize via capture of valence-band electrons: $A^0 + e^- \rightarrow A^-$. The ionization process is less efficient than donors because the density of states in the valence band is typically larger, and ionization energies are often deeper. At 150 K, we expect significant freeze-out.

Problem Statement

A p-type sample has $N_A = 10^{17} \text{ cm}^{-3}$, with $E_A - E_v = 0.045 \text{ eV}$. At $T = 150 \text{ K}$, estimate the ionized fraction N_A^-/N_A . Use $N_v(T) = 1.04 \times 10^{19} (T/300)^{3/2} \text{ cm}^{-3}$ and degeneracy factor $g_A = 4$.

Detailed Solution

Physical background: The fraction of ionized (negatively charged) acceptors is:

$$\frac{N_A^-}{N_A} = \frac{1}{1 + g_A e^{(E_F - E_A)/k_B T}}$$

The degeneracy factor for acceptors is typically $g_A = 4$ (accounting for the fourfold degeneracy of the hole ground state in the valence band).

Step 1: Compute Thermal Energy and Effective DOS

Thermal energy:

$$k_B T = 0.0259 \text{ eV} \times \frac{150}{300} = 0.0129 \text{ eV}$$

Effective DOS (valence band) at 150 K:

$$N_v(150 \text{ K}) = 1.04 \times 10^{19} \left(\frac{150}{300} \right)^{3/2} = 1.04 \times 10^{19} \times 0.353 \approx 3.68 \times 10^{18} \text{ cm}^{-3}$$

Observation: At 150 K, both N_c and N_v are significantly reduced compared to 300 K, making carrier generation more difficult.

Step 2: Charge Neutrality for p-type

Statement: With no donors:

$$p = N_A^-$$

Physical meaning: Every ionized acceptor creates a hole in the valence band. The hole concentration equals the ionized acceptor concentration.

Step 3: Hole Concentration from Band Statistics

Formula:

$$p = N_v \exp \left(-\frac{E_F - E_v}{k_B T} \right)$$

Define $\Delta_p = E_F - E_v$ (Fermi level height above valence band edge). Then:

$$p = N_v e^{-\Delta_p/k_B T}$$

Step 4: Acceptor Ionization Statistics

Formula:

$$N_A^- = \frac{N_A}{1 + g_A e^{(E_F - E_A)/k_B T}}$$

Rewrite in terms of Δ_p : Since $E_F - E_A = E_F - E_v + E_v - E_A = \Delta_p - (E_A - E_v)$, with $E_A - E_v = 0.045 \text{ eV}$:

$$N_A^- = \frac{N_A}{1 + 4 \exp \left[\frac{\Delta_p - (E_A - E_v)}{k_B T} \right]} = \frac{N_A}{1 + 4 e^{(\Delta_p - 0.045)/k_B T}}$$

Physical meaning: As Δ_p increases (Fermi level moves away from valence band), acceptor ionization decreases freeze-out sets in.

Step 5: Solve the Coupled Equations

System to solve:

$$N_v e^{-\Delta_p/k_B T} = \frac{N_A}{1 + 4 e^{(\Delta_p - 0.045)/k_B T}}$$

Substitute values ($k_B T = 0.0129 \text{ eV}$):

$$3.68 \times 10^{18} \times e^{-\Delta_p/0.0129} = \frac{10^{17}}{1 + 4 \exp \left[\frac{\Delta_p - 0.045}{0.0129} \right]}$$

Solution method: Again use fixed-point iteration, starting from an initial guess $\Delta_p \approx 0.02 \text{ eV}$.

Numerical result: The iteration converges to approximately:

$$\Delta_p \approx 0.037 \text{ eV}$$

Step 6: Compute Ionized Fraction

$$\begin{aligned} N_A^- &= \frac{10^{17}}{1 + 4 \exp \left[\frac{0.037 - 0.045}{0.0129} \right]} = \frac{10^{17}}{1 + 4e^{-0.62}} \\ &= \frac{10^{17}}{1 + 4 \times 0.538} = \frac{10^{17}}{1 + 2.152} = \frac{10^{17}}{3.152} \end{aligned}$$

$$\boxed{\frac{N_A^-}{N_A} \approx 0.20 \text{ to } 0.25}$$

Interpretation: At 150 K, only 20–25% of acceptors are ionized. This is a dramatic drop from the extrinsic regime at 300 K. The large ionization energy ($E_A - E_v = 0.045$ eV, nearly twice that of donors in Problem 7) combined with low temperature causes *severe freeze-out*. In this regime:

- Hole concentration plummets as acceptors neutralize.
- Device carrier concentration and conductivity drop sharply.
- This is the onset of the *freeze-out regime*, where device performance degrades.

Comparison with Problem 7: Donors freeze out at lower temperature than acceptors with equal ionization energy because of their lower degeneracy factor ($g_D = 1/2$ vs. $g_A = 4$).

Problem 9: Minority Carriers in n-type Silicon (300 K)

Context: In semiconductors, even when one carrier type dominates (majority carriers), the opposite type still exists at a reduced concentration (minority carriers). Understanding minority carrier concentration is critical for device physics: they determine reverse saturation current, leakage, and recombination.

Problem Statement

For $N_D = 10^{16} \text{ cm}^{-3}$ and $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, find the hole concentration p in n-type silicon at 300 K.

Detailed Solution

Physical principle: Mass Action Law In *any* semiconductor (doped or intrinsic), electrons and holes satisfy:

$$n \times p = n_i^2$$

This is a fundamental consequence of equilibrium thermodynamics and the Fermi-Dirac distribution. It holds regardless of doping.

Derivation sketch: From band statistics:

$$n = N_c e^{-(E_c - E_F)/k_B T}$$

$$p = N_v e^{-(E_F - E_v)/k_B T}$$

Multiplying:

$$np = N_c N_v e^{-(E_c - E_v)/k_B T} = N_c N_v e^{-E_g/k_B T} = n_i^2$$

where the last equality defines n_i .

Determine Majority Carrier Concentration

Charge neutrality for n-type (donors only, acceptors negligible):

$$n = N_D^+ + p \approx N_D^+$$

At 300 K with moderate doping ($N_D = 10^{16}$), donors are fully ionized, so:

$$n \approx N_D = 10^{16} \text{ cm}^{-3}$$

Compute Minority Hole Concentration

Apply mass action law:

$$p = \frac{n_i^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{16}}$$

$$= \frac{2.25 \times 10^{20}}{10^{16}} = 2.25 \times 10^4 \text{ cm}^{-3}$$

$$\boxed{p \approx 2.25 \times 10^4 \text{ cm}^{-3}}$$

Physical Interpretation

Ratio of concentrations:

$$\frac{n}{p} = \frac{10^{16}}{2.25 \times 10^4} \approx 4.4 \times 10^{11}$$

Electrons dominate holes by a factor of $\sim 10^{11}$. However, despite their tiny concentration, minority carriers are *not* negligible for device performance:

- **Reverse saturation current:** Dominated by minority carrier diffusion and recombination.
- **Switching speed:** Determined by minority carrier lifetime.
- **Leakage:** In reverse bias, minority carrier generation sets the leakage current.

Key insight: The mass action law ensures that as majority carrier concentration increases (higher doping), *minority carrier concentration decreases*. This inverse relationship is fundamental to all semiconductor devices.

Problem 10: Compensated Semiconductor and Near-Intrinsic Behavior

Context: Real semiconductors often contain *both* donors and acceptors. When present in comparable amounts, they compensate each other. The net doping determines the material type and carrier concentration. When compensation is nearly complete, the material behaves nearly intrinsic despite having high total impurity concentration.

Problem Statement

A Si sample at 300 K has $N_D = N_A = 5 \times 10^{16} \text{ cm}^{-3}$. What is the majority carrier type and minority carrier concentration?

Detailed Solution

Charge Neutrality in a Compensated Semiconductor

General statement:

$$n + N_A^- = p + N_D^+$$

where N_D^+ and N_A^- are ionized donor and acceptor concentrations.

For our case (complete ionization at 300 K):

$$n + N_A = p + N_D$$

$$n - p = N_D - N_A = 5 \times 10^{16} - 5 \times 10^{16} = 0$$

Therefore:

$$n = p$$

Apply Mass Action Law

With $n = p$:

$$n^2 = n_i^2 \implies n = p = n_i$$

$$n = p = 1.5 \times 10^{10} \text{ cm}^{-3}$$

Result:

Material is intrinsic: $n \approx p \approx n_i \approx 1.5 \times 10^{10} \text{ cm}^{-3}$

Majority Carrier Type: Indeterminate

Physical situation: With exact compensation ($N_D = N_A$), there is *no majority carrier* the material is electrically neutral and behaves like intrinsic Si. In real samples:

- A tiny imbalance $|N_D - N_A| = \delta$ (unavoidable in practice) sets the type.
- If $N_D > N_A$ by δ : Material is weakly n-type with $n \approx \delta$ and $p = n_i^2/\delta$.
- If $N_A > N_D$ by δ : Material is weakly p-type with $p \approx \delta$ and $n = n_i^2/\delta$.

Why Perfect Compensation is Rare

Device implications:

- Compensated semiconductors are used to create semi-insulating material (e.g., GaAs semi-insulating substrates).
- Exact compensation requires extraordinary control during crystal growth.
- Even 0.1% imbalance sets strong type and majority carrier concentration.

Example: If $N_D = 5.1 \times 10^{16}$ and $N_A = 5.0 \times 10^{16}$:

$$\begin{aligned}\delta &= N_D - N_A = 10^{15} \text{ cm}^{-3} \\ n &\approx \delta = 10^{15} \text{ cm}^{-3} \quad (\text{n-type majority}) \\ p &= \frac{n_i^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{15}} = 2.25 \times 10^5 \text{ cm}^{-3} \quad (\text{minority})\end{aligned}$$

Summary: Key Insights from Problems 7–10

1. **Impurity ionization:** Controlled by competing effects: thermal energy ($k_B T$), ionization energy ($E_c - E_D$ or $E_A - E_v$), and degeneracy factor (g_D or g_A).
2. **Freeze-out regime:** At low T , most impurities become neutral, severely reducing carrier concentration. This is a major limitation for cryogenic semiconductor devices.
3. **Temperature scaling:**
 - $N_c, N_v \propto T^{3/2}$ (DOS decreases with lower T).
 - Ionization fractions fall exponentially as T decreases.
4. **Charge neutrality + statistics:** The coupled system of equations (band statistics + impurity statistics + charge neutrality) must be solved *simultaneously*. Fixed-point iteration is a robust numerical method.
5. **Mass action law:** $np = n_i^2$ holds in equilibrium regardless of doping. This inverse relationship is central to all semiconductor physics.
6. **Majority and minority carriers:** Doping controls majority carrier concentration; the mass action law determines minority carriers. Both are essential for device operation.
7. **Compensation:** When $N_D \approx N_A$, the material behaves nearly intrinsic. Perfect compensation is rare but used intentionally for semi-insulating material.
8. **Degeneracy factor:** For acceptors, $g_A = 4$ (fourfold hole degeneracy). For donors, $g_D = 1/2$ (due to specific orbital/spin structure). This affects relative ionization efficiency.

Numerical Methods Note

For coupled ionization-charge neutrality problems, a simple *fixed-point iteration* works well:

1. Choose initial guess for Fermi level (or equivalently, $\Delta = E_c - E_F$).
2. Compute electron/hole concentration from band statistics.
3. Compute ionized impurity concentration from statistics.
4. Check if charge neutrality is satisfied: $n = N_D^+$ (or $p = N_A^-$ for p-type).
5. Update estimate of Fermi level and repeat until convergence ($|\Delta n| < \epsilon$).

Convergence typically occurs in 3–10 iterations for well-posed problems.